Filament deposition Manufacturing (3D printing) of definitely useful things

The zine for the workshop, by OpenERV

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This document as a google doc:

This doc as a pdf in a github repository:

Intro

Clearly, being able to make parts is useful, we are surrounded by machines etc that are in turn made partly of mechanical parts. Metal and other materials would be nice, but plastic is a start.

3d printing doesn't seem like the biggest deal until you realize how it changes the economics. The moment you really need some part, and try to get parts made by other people it's unbelievably expensive in many different ways even aside from money, compared with the actual work time and cost to make them once you are all set up, and even the setup cost/work.

- This is going to assume you are looking at buying a printer for yourself or your community space. Unfortunately I have found that attempting to use printers in libraries, makerspaces etc. is generally pretty impractical due to the time it takes to print things and the limitations they place on you. This may also be useful if you get your stuff printed by a service, or even want to open a service, but in general I think most of the value lies in people doing it internally to an organization and community. Printing stuff for each other is a good idea, but a formalized print service is a lot less efficient.
- I recommend you **ask not what printing can do but what you really want to do and how printing can help.**
- **Printing spoons or plates or chairs or whatever everyday things is IMO generally not that interesting.** Everyone tries to do this, to look around and ask what they might print, and how that might save money etc. The useful stuff is scientific experiments, custom stuff, development of products (which may actually be produced using another process eventually) and repair or retrofit of equipment, none of which we tend to see around us. It's the long tail of rare but unusually valuable uses where it shines.
    - **Getting an idea of what you can and cannot do is important, once you have that you will see uses for it.** Learning/getting set up is foundation building. You should have some idea of your primary use for FDM before investing, though. Personally, I knew I wanted to make machines, and at least plastic parts are better than nothing. Then I found it surprisingly useful. I used to be all about CNC milling. I would quite like to have a milling machine too, but it's just too expensive/space consuming etc. for what I would use it for. Printing has a better ratio. However it's true that not having a milling machine has warped what I work on a lot.
    - If you are doing something in the physical world that involves building stuff it can often play a very important and even critical enabling role. There will still be lots more to any interesting project, though, usually.
- Also I strongly recommend you don't just jump into something that seems cool. Make sure it's really worth doing, and you should assume it will take many tries to get a truly good result. Consider the maximum theoretical potential if things go well. Also is it really the *best* thing you could be working on? There is far too strong a tendency in the technical community to jump on cool things and then never get anywhere with them because it's not worth doing.

Fundamentals of the FDM process.

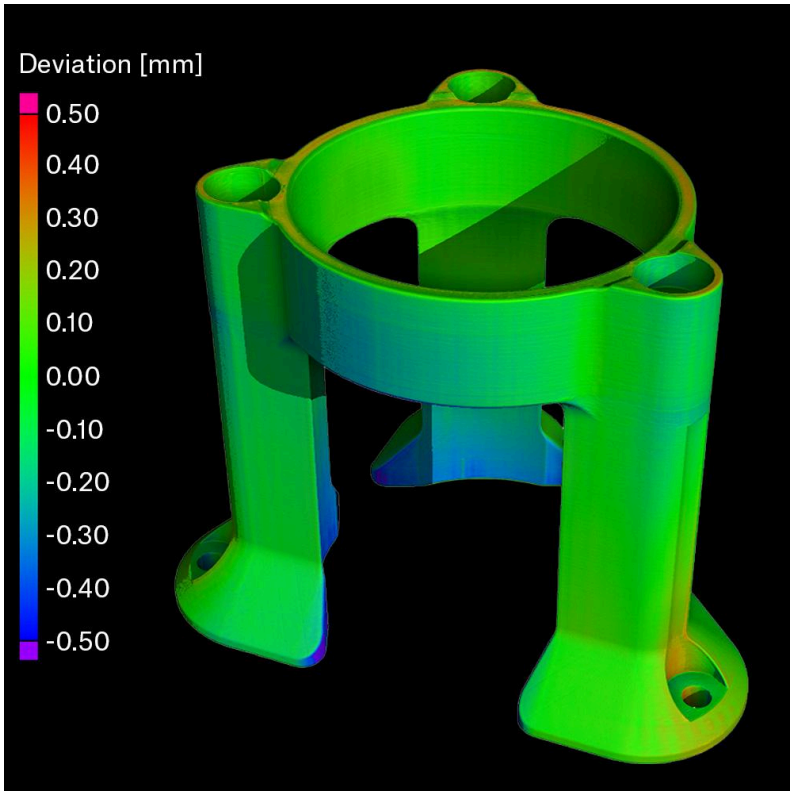
- Most fundamental:
    - **This zine is about accuracy, geometry and material properties and producing parts with low cost of labor, material, and capital.** If the color matters, it's not usually my department. All manufacturing processes have strengths and weaknesses, people claiming 3d printing can make "anything" are very silly.
        - **Material properties**:
            - Heat resistance, uv resistance, water resistance,
            - strength of course, mostly yield stress
            - hardness (rubber can be just the thing sometimes), this is usually measured in shore durometer A and D scales, for rubber and plastics. It is similar to rigidity, which is basically the same as modulus of elasticity.
                - **Stress** and **strain** are difficult concepts but how central they are cannot be overstated. **Stress** is the degree to which atoms or molecules of a material are too far apart or close together in a material, and which axis along which they are stretched or compressed, basically. So there is a magnitude and direction. **Strain** is the distortion that results from stress. **Internal stress** is one region of a part pulling or pushing on other regions of the same part for some reason (basically not enough atoms/molecules in some volumes of space vs others, compared to the ideal). It leads to distortion of parts, mostly, in this context, like elephant footing and bed delamination, this is important and you will definitely notice this and have to account for it plenty. **External stress,** or just stress, is imposed from the outside on the part, which is met by the reactionary stress that arises inside the part. Changes in stress generally lead to changes in strain and vice versa.
            - Wacky phenomena like the stress cracking PLA does
            - impact strength is also important. This is basically the amount of energy a sample of material can absorb while being hit by a hammer before it breaks. Some materials are very strong but have poor impact strength, rubber in contrast can absorb tremendous energy but isn't that strong. If a material has low impact strength then dropping the part made from it would not go well, for instance. Think a glass vs plastic plate.

- In-between is layer bonding, that's part-level rather than inherent to the material but doesn't fall into accuracy.

■ Accuracy

- **is how closely the object resembles the original digital description.**

- The picture below shows what is often dubbed a 3d surface deviation map or heatmap. The blue area on the bottom of the legs and the red at the "toes" of the legs indicates the legs are splayed out sideways considerably. You could put a triangular sheet of plastic on the feet to avoid most of this distortion, but the part would still be "trying" to distort, so the legs and the rest of the structure would bend slightly. You'd be able to see this in a sufficiently accurate and high resolution map. These maps are not very common and are expensive to make from real objects, a scanner can do deviations of this size but if you want to measure small fractions of a millimeter you need a pretty nice bit of equipment to do that. Usually we just measure it with calipers where things matter most, design things so that it doesn't matter much how the part distorts etc.



- This includes various minor defects, stringing etc,
- But mostly surface finish, and tolerances of dimensions
    - Tolerances can include the linear distances between two points, but also flatness, perpendicularity, roundness, straightness are particularly important. There are a lot of ways that are not merely random that parts and their features can vary from the ideal. We can break down the

error into different types, but we can also just talk about overall error.
        - Surface roughness. Dimensions refer to the average or outermost surface, despite the layer line ridges that appear in FDM parts. Roughness is the average small scale deviation from the average surface basically. It's all about different scales, roughness is just small scale inaccuracy basically, and in this case it has a certain repeating, fairly consistent nature i.e. it is not random. Roughness can be measured at different scales.
    - Balance and meeting requirements is actually the challenge here, not pushing the limit of accuracy at undu expense. Improving the ratio of quality to cost can't really go far wrong though.

■ Geometry

- Thin walls can be difficult or impossible
- Small areas that cannot be supported can be nearly impossible.
- In many situations it can be impractical to remove support material in some geometries. There are solutions like soluble supports in many cases, but most of us won't have that.
- Leaders. A "leader" is like the tip of a candy cane, if you hold it upright and imagine it forming from bottom to top, the tip of the hook would come to exist before it is connected to the rest. Obviously it needs to be supported against gravity when that happens, using support material. Some leaders such as a pointy tip, can be a problem as they have no area to connect to the support material. You can arrange extra support material where desired in the slicer to accommodate situations like this. The underside, where the support is located, tends to get messed up in various ways, this can be a problem.
- Wide vertical walls can if sufficiently thick be impossible due to contraction causing bed delamination even with the best countermeasures, also depends on polymer of course
- Some other geometries and with higher infills can be impossible due to bed delamination
- High overhang angles. It's possible to print higher angles than usually assumed, with certain techniques, but they are a real hassle. Recently developed techniques call allow very high overhangs but they are finicky, requiring precise thermal control that needs to be dialed in etc.
- Combinations of these can be nearly or actually impossible to print.
- Total size, of course.

- The tug of war stress accumulation and its implications: As each layer is printed, it ends up in a contractile state after deposition. It pulls on the region below.

Layer after layer pulls on the ones below. **This is a major problem in additive manufacture**.

- The main defence is to hold the part to the bed really well and try to avoid designs that have high vertical walls and other factors that can make this break through into a serious practical problem.
- Heated build volumes also can mostly solve the problem but only at sufficiently high temperatures and the current generation of machines is not well designed (they try to just bolt the functionality on after designing the machine, basically just heating up the whole machine, which causes a variety of serious problems). It also leads to the part warping even if the print is successful. It's worse with some materials than others, in general the higher the melting point (it's more of a continuum than a point for polymers), the worse this problem, but there are several factors that combine, such as coefficient of thermal expansion and even the room temperature.

The picture shows the stress in a simulated small printed plate on the bed. The stress in the middle is the highest because the material is not able to move as much. At the edges it can move, not being attached to as much else. At the lower corners it's attached to the print bed, again constraining it, so it's red there. Clearly after being released from the bed, the corners would sproing up, the bottom of the part would become slightly convex. The forces here can become strong enough to cause the part to detach from the bed during printing, ruining the print.

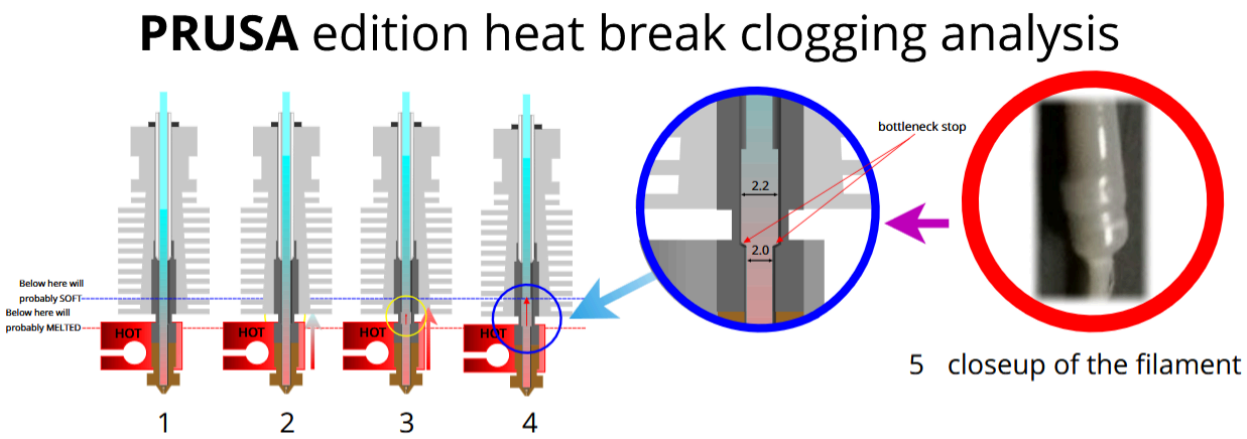
Need pic

- Higher infills give more force
- High vertical walls are a problem, the thicker the worse.
- It's a fight between this force and bed and layer adhesion, the bed itself will also bend some. Any actual motion will tend to be amplified as it gets baked in with the next layer.
- It can also cause serious warping even if the bed adhesion doesn't fail
- If the part can warp or starts to come off the bed during printing it gets even worse and things snowball. It's best if it's constrained, but even the bed is significantly flexible. Glass plates help.
- Compensation digitally is generally not done, the software tools are not ready for that.

- Layer adhesion can approach 100% theoretically, and in reality it can be pretty close, actually.
- High overhangs without support can be impossible or impractical, and support messes up overhangs.
  - Can get drooping or process failure if outside process window. The **process window** is the region where all process parameters like speed, temperature of the nozzle, ambient temperature, cooling airflow and so on are all in range to give an acceptable result. Their acceptable ranges can

be dependent on the others. If you think of only two parameters, say nozzle speed and temperatures, you can make a graph and draw a region on the graph that's ok, or there may even be several regions. Same deal with more parameters, just not as easy to visualize.
        - FDM is not force-free, the forces can cause problems with some overhang geometry, or thin tall geometry.
    - Shape of the part and how it deviates from the ideal
        - Bulging of the road at the perimeter (esp with inside out walls)
        - Messed up surfaces on areas support was used
        - Warping due to tug of war contraction
        - Wandering of the road/lack of consistency at the edges (constrains the layer thickness for a given nozzle diameter, and it constrains the process window to a certain temperature range)
        - Poor surface finish (layer lines, mostly)
        - Less so:
            - Many factors are not accounted for, like the variation of the road edge position with different layer heights, even the elasticity of the machine. These are arguably less fundamental though so they go in the next section.
- Less fundamental
    - Machine runs G-code
        - Gcode is a simple way to describe the plan for what the machine will do, it's not really a programming language but it does have some program flow control features like conditional statements and variables etc. but they are rarely used in 3d printers except during homing sequences etc. The actual printing process is mostly just a ton of commands to move between the endpoints of straight lines, and some changes in the speed the head should move etc. Acceleration is implemented by the computer in the machine that interprets the gcode, usually.
        - The cheap machine has very little sensors or input about what's happening out there in the build volume, it is pretty dumb. This presents a serious inability to recover from issues, all previous layers have to go ok. There is a limited ability to recover from small errors just because of how the process goes even without any brains, can snowball (some defects in thin walls for instance, or bed delamination).
            - Ai camera would be a good idea to at least pause the print in case of issue, Klipper has provisions for some stuff like that but I haven't tried it yet. But you can't go back, by the time you see there is an issue the part is usually scrap and you have to figure things out, modify the program and just try again. Remember this when you consider the costs of printing.
    - Poor quality of most machines
        - Lack of bed flatness. Glass beds help but you may not actually need to do that.

        - Lack of perpendicularity in the xy, xz or yz planes.
        - Lack of circularity, even compared with other 3d cartesian machines
        - Backlash, acceleration can be a bit of an issue but usually not too bad for us.
        - Twist, this is particularly hard to deal with. It's where one side of the machine has different perpendicularity error than the other. Basically the z axis is slightly twisted.
    - Fundamental limits on speed
        - At the nozzle tip
            - Plastic melts, gets pushed out
                - Viscosity is the main limit on flow rate, and the barrelling and force the extruder can produce before it slips. Sometimes there are other issues which are encountered as flow is increased.
                - Higher temperature is lower viscosity but there are limits as the more volatile fraction of the polymer blend can boil out, also it can stick to the trailing edge of the nozzle, or other issues can arise.
                - There are additives that can reduce viscosity, giving "high speed" filaments, but there are actually downsides, in my experience the process window tends to be smaller which can be a real problem.
        - So called heat creep in hot end, not as bad with more recent hot ends



            - Force of extrusion, temperature in the region above the really hot part of the nozzle (there has to be a region with some thermal gradient) equals barrelling and excessive friction. This can appear to cause a "clog" but it's not really the nozzle clogging, actual nozzle clogs are very rare in my experience, I suspect most are actually this.
            - If fan stops working or the thermal connection to the heat sink in the hot end is poor this happens at (much) lower extrusion forces.
    - Ringing and acceleration/synchronization defects.

        - Ringing is basically unimportant for us, it is a minor nuisance. Input shaping is the recently developed solution but imo that's a cheap out thing much like the auto bed levelling is a poor substitute for a machine that's actually physically good to go. Actual damping could have been used.
    - Nozzle leakage, backlash and pressure delay
        - Pressure delay is a result of the springiness along the path from the extruder to the actual nozzle. When the extruder starts moving the extrusion only ramps up over a short time period, rather than following the extruder motion perfectly. It also causes the extrusion rate to be out of synch when the nozzle accelerates/deaccelerates. So called pressure advance can mostly compensate for this. The springiness also causes the nozzle to leak during pauses and transport moves unless the filament is retracted. Obviously it needs to deretract it after too. The cheap machines don't do a great job on this in their built in code but the code can be modified, retract/deretract during actual printing is the slicer's job and is easy to arrange for.
        - Small blobs and bulging in various areas arise for a variety of reasons, mostly nozzle leakage of various kinds.
        - Stringing, not generally a big deal for functional prints.
        - **But any plastic that escapes when it shouldn't leads to insufficient plastic where it should be, too, unless compensated for, and compensating just right is error prone and likely to lead to more problems than it solves. It's usually better to just reduce leakage.**
        - The so called z seam occurs at the beginning and end of the roads at the perimeter (the road is the track of plastic left by the nozzle). There are ways of reducing it or randomizing the location but it's not usually a big deal. It can be a region that is an indentation rather than a bulge, this can be preferred.
    - The way the direct extruder pulls on the filament roll and how this can be a problem if the force on the spurs is already approaching the limit or if the head transits fast and the force to pull the roll is too high for a second, the motors can have their force limits exceeded and that ruins the print. Or there can be other failures like filament tangling etc.
        - Steppers and skipped steps: The motors that move the axis and extrude the filament are almost always "stepper motors" or "stepping motors". These are essentially, in this case (a low noise driver), fed what is nearly a sine wave of electrical uh power, it's the voltage but it's current regulated to not exceed a certain amount of current. One cycle of the sine wave causes the motor to move 1/200 of a rotation in the usual motors used. However if the force on the motor is too high for the amount of current the motor is being subject too for even a single step, it will "skip" that step. On most machines the machine is completely oblivious to this. This obviously will lead to the nozzle not being in the right place, fucking up the print and conceivably causing other issues. "Servo motors" and

closed loop stepper motors generally do not have this problem, but an overload would still lead to issues of course.
        - Main defense is putting the filament up high in the center of the machine and not going too fast in general. There are many things like that, the machines are not actually well designed as they are, they need some help in many ways, as you will see. A proper complete machine would have this factored into the design.
        - Some kind of filament feeder that regulated the tension on the filament to a low level would be a good idea and allow very large rolls such as 10 kg rolls, which are available.
        - Bearings on the roll don't work very well as it leads to tangles. There has to be fast enough dissipation of rotational energy of the roll or it keeps on going when it gets pulled on, and the filament comes off the roll and gets everywhere, leading to jamming.
- Materials fundamentals:
    - Tensile strength, yield strength
        - **Tensile strength** is before it breaks, **yield** is before permanent deformation, we are mostly interested in yield strength. It's usually in Megapascals
    - Elastic modulus
        - Basically how flexible a material is. Stress vs. strain ratio. It's also in (Mega)pascals.
    - Shore durometer (elastomers)
        - Closely related to the flexibility/elastic modulus but tested differently.
        - Shore durometer is measured with an indenter, a probe of certain geometry is pressed into the sample with a certain force, the deeper it goes the lower the durometer basically. There are different scales which have different probe shapes and forces. D is for hard plastics, A is common for rubbers. 70D is like hard rubber for instance, 85D is typical 3d printed plastic. 95A is less hard rubber, 70A is too soft to work in a lot of printers so it's not that common, really soft rubbers like silicone gel are measured in O or other scales.
    - Impact strength
        - izod-charpy is the most common test. They hit a sample with a swinging hammer with higher and higher kinetic energy until it breaks.
    - Heat resistance, uv resistance
        - PETG is better than pla but it is NOT good to 75 deg C like people claim. ABS is better, PC and nylon are even better but harder to print partly due to their heat resistance. The brand also matters/exact formulation. Mfrs are tempted to reduce the heat deflection temperature (HDT) because it gives easier printing (less contraction), which people notice, and then they just lie about their HDT since most people won't notice/check.
    - Phenomena

- Creep. Polymers all exhibit creep, but the lower the heat deflection temperature the worse. This is a slow deformation that occurs over time when the material is under load. Carbon and glass fibers reduce it. It can be in a practical and non problematic/insignificant range in many situations though, it is noticeable with PLA/MPLA though in many contexts.
- **Stress cracking in PLA**
  - This is a wacky phenomena that leads to pla becoming fragile and brittle when exposed to stress. It is **NOT** the same as normal breakage due to excessive force. The material will slowly become brittle and weak in areas exposed to even moderate stress over a few months.
- Deterioration from water and oil (TPU tends to get fucked up by this, there are three major types of TPU. Polyester, polyether and polycarbonate. Guess which ones are more common and also which are more crappy, in general polyester is the most common and crappy, polyether is pretty good and is rare but obtainable through digikey etc. and polycarbonate is the best but I have never found any filament that uses it.)
  - Elastomers also are prone to a sort of creep but it's called "compression set".

Advice on which printer to get.

Let's face it, the main way to leverage FDM printing implies having a printer of your own. Don't forget you also will end up buying and storing extra filaments, spare parts, and some post processing stuff, maybe bed adhesive etc. though. Still for $600 CAD you can get all that put together, which would not last long if you buy printed parts from someone.

- Bowden vs direct extruder, direct is way better that's what you want.
- I only have experience with tronxy and a couple creality models, I have heard good things about prusa. For our purposes bambu is not recommended, they may seem to work for now but you will be confined and they will cut us all off soon enough.
- Klipper is important, don't settle for marlin
- Definitely you want the z-probe, all the good ones have them now.
- Pei build plate is optional, glass works pretty well and it is flatter, adhesion is good either with with bed adhesive. You can almost always change the plate later.
- Avoid tronxy, they are absurdly bad in every way
- Creality is ok, good price/performance compromise for me usually,and Prusa is probably better but much more expensive and I haven't actually tried it thusly. I haven't tried Elegoo, but they seem ok and make good filament at least. Anycubic and many others I have no idea. I avoided their SLA stuff as it doesn't sound very good, but they have learned something about FDM in all these years.

    - Don’t stretch to get a bigger print volume too much, 300 mm build plate would be pretty nice, but more than that, large prints are a real hassle anyway you’ll probably prefer to avoid them. There have been many times I wanted to upsize from the 220 mm build plates, but none that have actually been worth buying another printer for, or ordering parts from someone else, either.
    - Soluble supports I have wanted many times, and it doesn’t cost that much. I have next to no use for more than 2 materials.
    - Heated build chamber would be good but they don’t do a good job on it yet. They try to heat the whole machine, which is dumb as rocks. There needs to be a flexible walled chamber around only the build plate and nozzle, that I would buy. You can actually print PAHT and other PA blends with carbon fiber, without a heated build chamber, for some geometries, but heated build chamber is a good idea for sure.
        - You need more like 380 degrees nozzle though, not what they claim, for good layer adhesion, for higher temperature filaments like PAHT, and this is not very common. Many printers can actually be pushed to this level but better to get one that can do this stock of course, if you plan to do so. I’ve never seen a cheap one that advertises this.
    - The newer extruders like the unicorn one from creality really are legit a bit better and I would prefer to have that. The older ones are fragile, for instance, they can be easily damaged while replacing nozzles. Also the teflon tubes wear out, the unicorn doesn’t have that problem.
    - Remember to factor in how easy it is to get replacement parts, and their cost, you don’t want to get stuck with an unrepairable machine. You also want to be able to expand to hardened steel nozzles or whatever, they may not be available for less common printers. A good printer should be modular i.e. not a custom hot end, but one also shared by other printers at least in the same company. Likewise, it should use Klipper etc. not some wacky shit as they will never put the work in to do a good job.
    - Ball bearings are a lot better than the v-rail things, or bushings. Rail bearings are better than the rods.

Slicing

- We are basically planning the print process here, and then the slicer bakes our plan, then the post-processor makes it into gcode, and then a few things are added in various places, and the code is ready to view or test.
    - Therefore this is the place to include some discussion of the practical process parameters etc.
- The **process window** is an important concept you won’t see discussed in most stuff on 3d printing, because it comes from the industrial world. It’s the region where all process parameters like speed, temperature of the nozzle, ambient temperature, cooling airflow and so on are all in range to give an acceptable result. Their acceptable ranges can be dependent on the others. If you think of only two parameters, say nozzle speed and

temperatures, you can make a graph and draw a region on the graph that's ok, or there may even be several regions. Same deal with more parameters, just not as easy to visualize. It can be different for different areas for the same part even.
    - We can use "modifiers" in PrusaSlicer to get different process parameters in different areas of the part, but it's a hassle.
- I actually go through almost every single setting quickly, that helps a lot, unless I've saved the 3mf file and it is a very similar part. It takes a while to learn what each setting does, you can just leave it as default and explore how to correct/improve the toolpath with whatever settings you need as you go, though.
- Main slicing tricks for functional parts
    - Modifiers
    - Nozzle size
    - Variable layer height
    - Brims are important I almost always use them, the exception being many small parts or if you really don't need it (low z axis extents etc).
    - Make sure you name your gcode file sensibly
    - **Always save your 3mf file, sensible name**
    - Keep the file open, you may need to modify it and restart the print, that happens a lot. You can open multiple instances of prusa for multiple parts.
- Slicing checklist example, you will need to make your own.
    - Z-lift for taller parts?
    - Speed ok
    - Layer thickness ok
    - Road width/nozzle diameter
    - Retract ok
    - Outside in perimeters vs inside out ok
        - Outside in gives more accurate perimeters but in rare cases can lead to blobs in the center of posts that are all walls (material accumulates in the center).
    - Temperatures
        - I no longer change the bed temperature after the first layer, with the bed adhesive it doesn't help and I suspect it makes things worse due to differential contraction. But it does theoretically make some sense as you could get better bonding at higher temperatures and yet the mechanical strength of the plastic would be too low. So then you drop the temperature after the first layer to increase strength. It would be a fight between the detriment caused be differential contraction and this influence.
            - The print bed material (steel or glass usually) contracts at a different rate per degree C than the plastic. It is therefore very important that the interface temperature between them not change, as they have a strong tendency to break apart. But when the print is done this is very handy, just let the bed cool and the part comes off a lot easier.

        - Even 5-10 degrees in the nozzle temperature can have a beneficial or adverse effect, on stringing and even a serious effect on layer adhesion and strength of the polymer. One MPLA brand would start to bubble if it was only 10 degrees above optimal. This means the ambient temperature can have an impact. Even the radiant heat from the bed has been known to cause problems.
            - Different nozzle materials and flow rates can also have an effect, taking you out of the process window in some cases. Heat has to flow from the actual heating element through the metal of the nozzle and into the plastic. The higher the rate of extrusion/flow, the greater the drop in temperature along that path. Thus the actual temperature of the plastic coming out the end will go down as the flow is increased, it can also start higher and then drop a bit after travel moves etc.
    - Brim ok? Usually needed.
    - Basically you should usually review all settings however briefly, just scan over them and think if that's ok or not.
    - View the gcode if in doubt or it's a tricky part, it's not always exactly the same as the preview. You can also use the gcode viewer to look at existing programs if you forget exactly what that gcode makes.
    - **Always save your 3mf files**

Machine setup upon first build or some types of changes/damage

- Install moonraker, this is a web interface/gui for Klipper, it should come installed but sometimes doesn't. Sometimes you can just enter the ip address of the printer plus tell it to go to port 4408 and there it is. So like http://192.168.4.24:4408 .
- Level the bed mechanically if you can
    - Needs to be adjustable, I modified mine.
- In general it's better to use a glass plate if you need it to be flatter than 0.4 mm deviation across the surface or so.
- Perpendicularity correction in klipper
    - Cannot usually be mechanically adjusted unfortunately
    - Print a YACS STL set for the xy plane, but beware it doesn't really work for the planes in Z because the bed is not flat or level (perpendicular to the Z axis or parallel to the XY plane the head moves in are also separate issues).
        - **The level and flatness of the bed is not the same as the perpendicularity of the axis to each other. They need to be treated separately.**
    - Find instructions online for measuring and using YACS and get the figures for xy
        - A better way to polish off the last bit of error is to print two identical squares, then hold them on top of each other, with one upside down and

one edge on a table. The side edges will not be aligned unless the perpendicularity is good. They will instead be at an angle to each other if the squares are skewed. This is more sensitive than the YACS approach. You just have to estimate the error in radians, try it and guess again. A binary search is good. So it takes a few rounds.

- Print the 3 walled object, corner of a hollow cube, 60 mm on a side is good.
  - Measure the top with a square then use a spreadsheet that takes estimated angular error using a square and gives the figures klipper needs using some trigonometry
- Put the following in the config file in Moonraker to get it to run the gcode to correct skew on each boot, obviously with your own skew correction figures:

```
[delayed_gcode my_delayed_gcode]
initial_duration: 1.0
gcode:
  G28
  SET_SKEW XY=141.59,141.77,100.16 XZ=141.22,141.17,100.16 YZ=141.7,141.03,100.16
```

- Verify it's working with the command line thing and by printing sample part
  - Use command GET_CURRENT_SKEW in the klipper console, in moonraker
- Measure the *top* and xy, can ust cut the corners off the yacs thing with a knife. More plastic on there would probably be good as it wobbles around a bit during th print.
- Adjust with method on command line in moonraker or whatever
- Can just use a square, the square hole and peg method might warrant mention as well, or comparison with one side against a table. But measure both sides, they should be the same when done.

- Twist of the Z axis is a problem, there is no good way to compensate for this, it would require compensating the stl itself which is a hassle and error-prone if the orientation of the part is not just right on the build plate. Clearly when we add this all up, a better machine is warranted for better accuracy in some cases.
- Machine adjustment
  - Installing moonraker etc. or finding the proper full-featured view
    - Such a long time ago I mostly forget but sometimes it's included, you just need to go to the right port, 4408 and then you get the full featured view, not the rinky dink clowny tune view that doesn't work.
  - Mechanical bed levelling
    - Mod for ender 3 v3
      - 3 points is ideal but more fragile
      - Can use 4 and actually use the bending to bend the plate back

        - Turn bed level off, print a brim very slowly and adjust as you go, easier.
            - How to turn bed levelling off again?
        - Need to use a block and nozzle, move it around and do most o fhte levelling first so it can even print the road though.
- Linear advance (quick note)
    - It's usually fine, best thing is to wait till it matters and adjust it then with an actual part that it matters for, or print a sample parts and improve it a bit
- Don't mess with flow, that's a common mistake, issues should be solved differently. Similarly linear motion rarely needs to be changed.
- PID tuning in Klipper, this is important for some printers.
    - Klipper has auto tune, I forget exactly how to us it now but it worked great.

Maintenance/operation every few prints:

- Use checklists at first, same as for slicing.
    - I use bed weld bed adhesive, the other ones are not quite as good, that I have tried. PETG doesn't seem to even benefit much from bed adhesive, on a PEI textured plate at least.
    - Might be a good idea to redo the nozzle offset if quite different nozzle temperature.
    - Do the bed levelling routine after any significant changes, usually it's not a good idea to do it every time.
    - Belt tightness, rarely an issue but needs to be checked on rare occasion
    - Grease the bearings and leadscrews, you really don't want them to run out of grease.
    - Maybe replace worn nozzle, it takes a long time for a nozzle to wear out with most filament but it can happen eventually and lead to issues if you don't watch. Be very careful and do the nozzle change right, if and when you do, I minimize the need to change it, it's easy to break it on many printers they didn't do the heat break very well, some of the recent extruders solve this problem.
        - Use two wrenches, squeeze the handles together so very little torque goes on the heat break.
    - If changing filament, then you take filament out, make sure the end of it doesn't sproing. That's what leads to tangles, it's when the end of the filament goes underneath a loop of filament. If you always keep hold of the end you won't get tangles. Put the end through the holes in the roll, and let lots of excess on the other side hang way out so it can't come out of the hole by accident.
    - The trick where you peel off the road as it's making the brim to determine adhesion directly is a good trick, not needed every time though.

Operation/maintenance for each print:

- Again, use a checklist:
    - Did you use the preview to make sure the slicing is good?
    - Got the right part?
    - Make machine is set up such that it's coherent with the slicer settings.
        - Correct filament?
        - Nozzle diameter?
        - Nozzle material is what you think?
    - "Calibrate" done? as needed, means bed levelling and nozzle offset, nothing else. It's not calibration but that's what they call it. May need to be redone if nozzle changed or bed different temperature this time.
        - Re-calibrate if nozzle changed or bed temperature much different.
    - Clean bed every time, no exceptions, or at least in the area you know the part is going.
        - I use a sponge bottle with water but that's with the adhesive, alcohol is good if you don't use bed adhesive, make sure there is no extra crap in it (the humectants.)
    - Start print, then watch the first layer for a bit while it does the brim or skirt
        - Watch the road, you get to know what a good road looks like for a given nozzle diameter. It looks different for different colors though which is another reason I don't like changing the color!
- During print
    - watch the first run of any new gcode usually, at least intermittently, or at least if you are trying something a bit sketch. If even just to save filament.
    - Watch the print to figure out how defects arise if/when they are a problem.
    - Listen for wayward noises
    - A part is usually scrap if something goes wrong. Some people try to fix things but it doesn't work very well and it's almost always better to solve things upstream in the gcode etc. and just try again. Sometimes I fix blobs that the head could run into, but I have been able to solve those in other ways anyway, it's too laborious to do that for a single plastic part in almost all cases it wouldn't be worth it.
    - Sometimes I let the print finish even if I know it's going to be scrap, to see if the rest of the program is ok/collect more info.
- You can move the machine manually when the motors are off, this is less annoying and less prone to mistakes in some ways
- After print:
    - Let the bed cool completely or nearly until removing parts, they come off a lot better and you are less likely to damage the machine as a result. Or you can take the magnetic plate off, I just don't bother. Even if you do, might as well let it cool. This is caused by differential thermal contraction, it is very hard for two materials to stay stuck together when the temperature changes and they have different coefficients of thermal expansion.

    - I use a dulled box cutter to remove parts, be very careful not to slip
        - Rock it back and forth, that helps a lot
        - There may be better tools, but the usual scrapers are not very good either.
        - Always cut away from yourself in case it slips!
- Quality control
    - You have to make sample parts and break them to verify good layer bonding. Different filament is often surprisingly different. It can even be affected by the nozzle material and ambient temperature.

Post processing

- Generally not a big deal
- Deburring tool/Box cutter to remove brim
- Dremel for blobs, not usually a big deal on small printer
- Remove support material.
- Drill holes to right size, they are usually too small. Trying to get them just right is usually going to lead to other issues and won't solve the defect caused by the start/end of the road anyway.

More general advice:

- Have a repertoire of materials, don't buy new/different stuff willy nilly.
- Each thing should have a generous margin against not working,
- Things should be repeatable so what worked before works again at least, that is what saves us. If it wasn't for repeatability none of this would be worth doing. This is why we use the same settings, and always saving the 3mf file, and always using procedures when cleaning the bed and prepping the machine, same materials too.
    - Repeatability is a foundation for reliability. It also makes things a lot easier to figure out, and less stressful in general. In a way all of manufacturing depends on repeatability to a remarkable degree. It takes a long time to figure things out and it wouldn't be worth doing if you couldn't use the result many many times into the future.
        - Being able to roll back to the previous version is also important. Not possible with a physical system but that's a good reason to save 3mf files. This can be important for CAD too, save as new file for major changes.
- Watch out for the phenomena where you change something in the process of fixing something and forget to put it back when it turns out to be something else, thus messing things up in the future.

    - Don’t mess with flow, that’s a common mistake, issues should be solved differently. Similarly linear motion rarely needs to be changed. The stepper motors make exactly 200 steps per rotation, the thread pitch and so on is also very precisely defined, so the flow and linearity are not the problem.

Materials and material properties

- MPLA. Modified Polylactic acid, generally significantly better than PLA, it is much less prone to the stress cracking problem, and considerably stronger.
- PETG, more heat resistant than MPLA but also more contraction.
- Elastomers
    - TPU, fiber reinforced TPU is stiffer and I suppose stronger?. Polyester TPU has issues with water and oil, there are other kinds but they are hard to find.
    - PEBA is more chemical and uv resistant, and easier to find than the more resistant TPU varieties.
    - That one tpu/nylon blend 72D was good stuff.
- PHA is worth mentioning but it’s pretty niche and not that useful. It’s fully biodegradable.
- PA6, PA12 and the wacky blends like PAHT
    - Elegoo PAHT-CF looks like you could do a lot with it even without a heated chamber but it takes more than 350 degrees nozzle, maybe 380 would be enough. Didn’t want to try it, might fry something.
    - Glass and carbon fiber reinforced varieties are important but the fibers cannot help much in the z axis. The increased rigidity does actually improve z axis strength when subject to bending but not of course pulling.
- Marginal utility:
    - ABS, not that bad actually, haven’t tried it much.
    - ASA Supposed to be largely like ABS but less contraction, definitely better than PCTG for most things I tried it for but in the end I stuck with PETG
    - PCTG, not apparently actually stronger or better than petg except it’s less stringy. I bought a bunch but could not get it to stick to the bed, fuck, tried two different brands, avoid this.
    - PLA, unmodified
        - Serious problem with stress embrittlement.
    - Haven’t been able to try
        - PC
        - PC blend from prusament, sounds cool but expensive and hard to obtain. Also the cf reinforced PC blend.
- DIY testing
    - Hot water and filament to compare, with IR or other thermometer, for heat resistance.
    - Print the parts and try to break/bend them as mentioned.

CAD

- CAD has to be part of the program because it's the custom stuff that's the most valuable.
- I would use onshape if I had to do it all over, fusion sucks. Constantly throws errors etc. for basic operations. You need a fairly powerful computer, the CPU is more important than the graphics card actually, most of the computation happens on the CPU, and multiple cores doesn't help that much, the main computations cannot be done in parallel.
- **Learning cad:**
    - Learn the basic tools first, and learn to solve challenges primarily with them, moving beyond only when needed.
    - Tools to learn in onshape or fusion first. You get a sort of combinatorial explosion, once you know the basics you can usually do whatever you want but it will take a while, beyond that the tools are mostly/usually about saving time or simplifying the workflow, or doing things like animations or making wacky shapes for hand grips or something which you rarely actually need etc.
        - Extrude (join, cut)
        - Boolean combines join, cut, intersect
        - Rotate
        - Push pull
        - Offset surface/face (solid body version)
        - Move face
        - Move body, with and without copy
        - Construction plane tools
            - Offset plane
            - Plane along path
            - Midplane
            - At least have a look at the other ones.
        - Loft
        - Section analysis
        - Sweep (not so often used)
        - Solid body Mirror (very important)
        - Sketching tools
            - Dimension
            - Mirror
            - Line
            - Circle
            - Rectangle tools
            - Arc
            - Spline
            - project/intersect
        - Split body, this one fucks up a lot but it's useful.
        - Coil (for threads and spirals)
        - Hole wizard and sketching to produce suitable points for it.

    - **Make a plan first before you start the actual cad.** The hard part is gaining enough experience to be able to even make a plan.
    - Don't try to use it for ideation or visualizing ideas, and don't modify it needlessly. Fork the file for major changes so you can roll back easily if you need to. Fusion has version management stuff but I don't use it as it doesn't seem very reliable, and you can't view things easily or move parts between versions etc.
    - Modifying the timeline is used sparingly, I just change the geometry directly usually, then every once in a while export everything to STEP or erase the timeline.
    - Major modifications or experiments should be in a separate file, then weave them back into the main CAD file only if they work out.
    - Remember that during the cad process you are doing a lot more than actual cad. You have to measure things in the real world, make all kinds of design decisions and so on. You have to budget time for all that. "Pure" cad is called drafting, it's when you already have a drawing on paper and just want to put it in the computer.

Other discussion

- A word on using ai for cad, the hype seems to be as usual greatly ahead of the technology but it may be a useful tool, but I haven't tried it much. It may be helpful for learning cad but I think you probably still have to know how to do things. The most promising paths seem to be OpenSCAD and similar python tools used with other scripts to drive geometry that you review and ask for changes etc. save things as STEP then go straight to STL and print when done. I don't know how you'd verify dimensions and clearances etc., I guess you'd open it in Fusion from STEP and just measure it. That may not be a whole lot faster for a lot of stuff, but it may have its place to get a project started, I guess.
- It would be a good idea to have some people in the community that are just relatively good with printing and they do collaborate with others. Why is this a promising idea while cloud printing is not? Crapitalism, basically. Crapitalists want only the easy jobs, and they don't actually know how to do it very well, and the technician only gets a tiny fraction of the money. The profit margins are too large. 3d printing is amenable to an individual with a machine accepting challenges from people in the community.
    - For example I deliberately asked around at foulab if I could help anyone with anything, and Marc-Andre said yeah I could use some drone parts, and I was able to help and the parts worked out great. That was really easy but the cloud services don't even offer the material we used, which worked considerably better. Also I just happened to be there, we just looked at things, drew some stuff on the computer and printed it right here, then I made some more stuff from different material in a a few minutes at home. It was really easy and took very little time but they would have charged way too much for all that put together. It would also have taken way longer, longer leadtime and probably more hours on Marc-andre's part. It's inherently more efficient and effective to have people in

our community that are relatively good at printing and to collaborate with them, than a cloud printing service is, given the nature of 3d printing and the projects that good uses of it entails.

- With cloud printing:
  - They make ridiculously bad tolerances as their promise. In reality the parts are usually a lot better than they promise but you can't know what to expect then, making things harder in some ways than just doing it yourself.
  - Long turnaround time
  - Expensive shipping
  - Cannot push geometries, materials
  - Cannot consult a person, the back and forth is useful.
  - However I have noticed even people who spend large amounts of time on 3d printing are bafflingly often not actually very good at it, there was two people I tried to offload some printing onto, and they really couldn't do even basic things, so don't do that. In general it's a technological competency problem in the world of atoms and manufacturing which is lacking in our north american society. I don't know why it happens but it does.